

Gavin Klorfine—Assign3

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DDAR Exercise 4.2

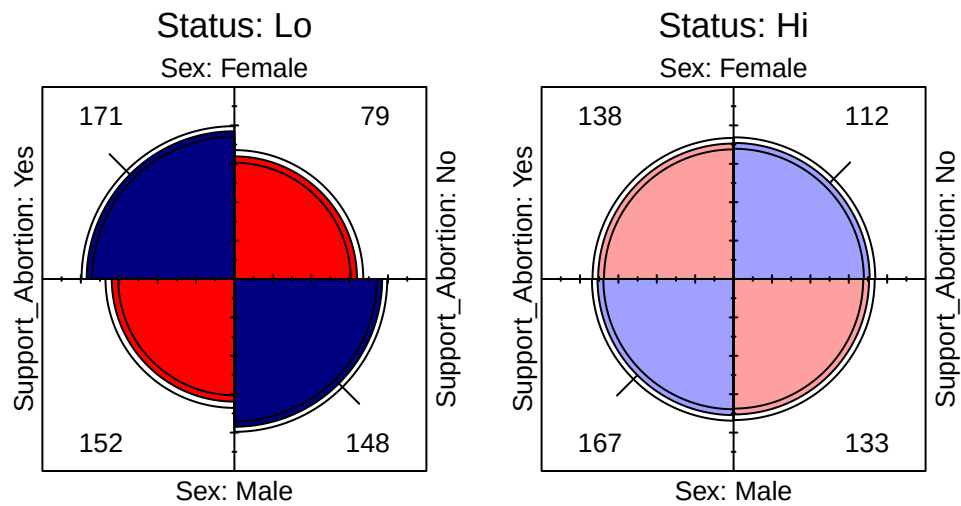
The data set Abortion in vcdExtra gives a $2 \times 2 \times 2$ table of opinions regarding abortion in relation to sex and status of the respondent. This table has the following structure:

```
data("Abortion", package = "vcdExtra")
str(Abortion)
```

```
## 'table' num [1:2, 1:2, 1:2] 171 152 138 167 79 148 112 133
## - attr(*, "dimnames")=List of 3
## ..$ Sex : chr [1:2] "Female" "Male"
## ..$ Status : chr [1:2] "Lo" "Hi"
## ..$ Support_Abortion: chr [1:2] "Yes" "No"
```

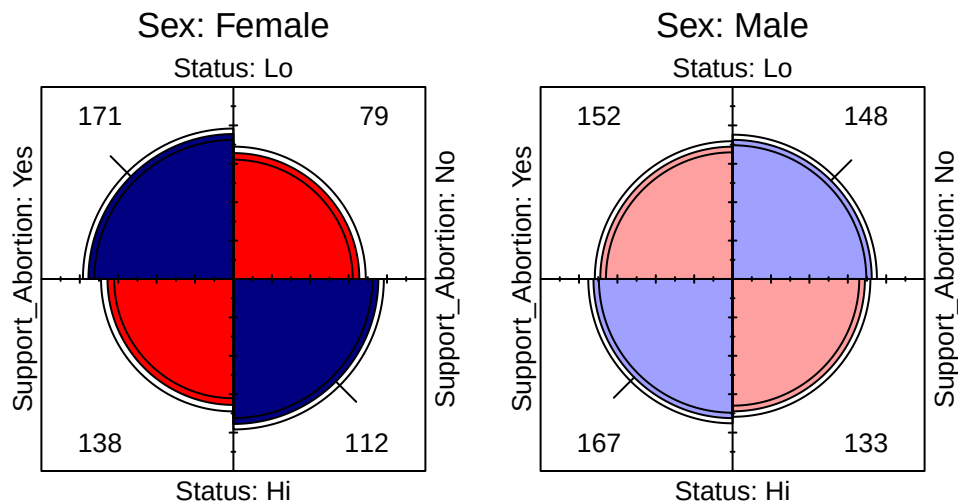
(a) *Taking support for abortion as the outcome variable, produce fourfold displays showing the association with sex, stratified by status.*

```
strat_status <- aperm(Abortion, c(1, 3, 2))
fourfold(strat_status)
```



(b) Do the same for the association of support for abortion with status, stratified by sex.

```
strat_sex <- aperm(Abortion, c(2, 3, 1))
fourfold(strat_sex)
```



(c) For each of the problems above, use `oddsratio()` to calculate the numerical values of the odds ratio, as stratified in the question.

```
oddsratio(strat_status, log = FALSE) # 4.2a
```

```
## odds ratios for Sex and Support_Abortion by Status
##
##      Lo      Hi
## 2.1075949 0.9812874
```

```
oddsratio(strat_sex, log = FALSE) # 4.2b
```

```
## odds ratios for Status and Support_Abortion by Sex
##
##      Female      Male
## 1.7567419 0.8179317
```

(d) Write a brief summary of how support for abortion depends on sex and status.

In the fourfold plots stratified by status, when looking at low status individuals, we see that support for abortion depended on one's gender, with odds ratio significantly greater than one. Moreover, low status individuals were more likely to support abortion, whereas high status individuals were more likely to be

against abortion. In contrast, support for abortion did not depend on sex when individuals were of high status, with odds ratio not significantly different from one.

In the fourfold plots stratified by gender, when looking at females, support for abortion depended on one's status, with odds ratio significantly greater than one. Moreover, low status females were more likely to support abortion, whereas high status females were more likely to be against abortion. As for males, support for abortion did not depend on status, with odds ratio not significantly different from one.

DDAR Exercise 4.4

The *Hospital* data in *vcd* gives a 3×3 table relating the length of stay (in years) of 132 long-term schizophrenic patients in two London mental hospitals with the frequency of visits by family and friends.

(a) Carry out a χ^2 test for association between the two variables

```
data("Hospital")
chisq.test(Hospital)

##
##  Pearson's Chi-squared test
##
## data:  Hospital
## X-squared = 35.171, df = 4, p-value = 4.284e-07
```

(b) Use *assocstats()* to compute association statistics. How would you describe the strength of association here?

```
(hosp_stat <- assocstats(Hospital))

##
##              X^2 df    P(> X^2)
## Likelihood Ratio 38.353  4 9.4755e-08
## Pearson          35.171  4 4.2842e-07
##
## Phi-Coefficient   : NA
## Contingency Coeff.: 0.459
## Cramer's V        : 0.365
```

The association was significant ($\chi^2(4) = 35.171$, $p < .001$), with Cramer's $V = 0.365$ indicating the strength of association was substantial.

(c) Produce an association plot for these data, with visit frequency as the vertical variable. Describe the pattern of the relation you see here.

```
assoc(Hospital, shade = TRUE)
```

